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Location-based determinations

Abstract

A method for location-dependent determination comprises: determining, with one or more processors, a geographic location of a user's electronic device using first data; in response to determining the determined geographic location is insufficient for performing a location-dependent determination, requesting second data to determine an updated geographic location of the user's electronic device; performing, with an Application Programming Interface (API), the location-dependent determination based on the updated geographic location; and completing, with the API, a transaction using the location-dependent determination.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority as a continuation of U.S. application Ser. No. 18/336,657, filed Jun. 16, 2023, which claims the benefit of priority as a continuation of U.S. application Ser. No. 17/181,494, filed Feb. 22, 2021, the entirety of each of which is incorporated by reference herein.

FIELD

(1) The subject matter disclosed herein generally relates to the technical field of special-purpose machines that determine the location of a user's device. Once the user's device's location is determined, determinations that are location-dependent can be made.

BACKGROUND

(2) The present subject matter seeks to address technical problems existing in making determinations based on a user's location. For example, an IP address of a user's device may indicate a location of an internet service provider's or mobile carrier's gateway and not the location of the user device. Accordingly, a location-dependent determination may be inaccurate due to the inaccuracy in determined location.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) Some embodiments are illustrated by way of example and not limitation in the views of the accompanying drawing:

(2) FIG. 1 is a block diagram illustrating a networked system, according to some example embodiments.

(3) FIG. 2 is a block diagram showing architectural aspects of a publication system, according to some example embodiments.

(4) FIG. 3 is a block diagram illustrating a representative software architecture, which may be used in conjunction with various hardware architectures herein described

- (5) FIG. 4 is a block diagram illustrating components of a machine, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein.
- (6) FIG. 5 is a block diagram showing aspects of an online method for conducting a transaction between a merchant site and an electronic user device using a payment processor, according to some example embodiments.
- (7) FIG. 6 is a block diagram illustrating a system, according to some example embodiments.
- (8) FIG. 7 is a flow chart illustrating a method in accordance, according to some example embodiments.
- (9) FIG. 8A and FIG. 8B illustrate a graphical user interface in accordance with some embodiments.

DETAILED DESCRIPTION

- (10) “Carrier Signal”, in this context, refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by a machine, and includes digital or analog communication signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device and using any one of a number of well-known transfer protocols.
- (11) “Client Device” or “Electronic Device”, in this context, refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistant (PDA), smart phone, tablet, ultra-book, netbook, laptop, multi-processor system, microprocessor-based or programmable consumer electronic system, game console, set-top box, or any other communication device that a user may use to access a network.
- (12) “Customer's Electronic Device” or “Electronic User Device”, in this context, refers to a client device that a customer uses to interact with a merchant. Examples of this device include a desktop computer, a laptop computer, a mobile device (e.g., smartphone, tablet), and a game console. The customer's electronic device may interact with the merchant via a browser application that executes on the customer's electronic device or via a native app installed onto the customer's electronic device. The client-side application executes on the customer's electronic device.
- (13) “Communications Network”, in this context, refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network, and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or another type of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High-Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long-Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.
- (14) “Component”, in this context, refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, application programming interfaces (APIs), or other technologies that provide for the partitioning or modularization of particular processing or

control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components.

(15) A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors.

(16) It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instant in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instant of time and to constitute a different hardware component at a different instant of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In embodiments in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

(17) The various operations of example methods described herein may be performed, at least

partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented components may be distributed across a number of geographic locations.

(18) “Machine-Readable Medium” in this context refers to a component, device, or other tangible medium able to store instructions and data temporarily or permanently and may include, but not be limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache memory, other types of storage (e.g., erasable programmable read-only memory (EPROM)), and/or any suitable combination thereof. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing instructions (e.g., code) for execution by a machine, such that the instructions, when executed by one or more processors of the machine, cause the machine to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

(19) “Processor”, in one context, refers to any circuit or virtual circuit (a physical circuit emulated by logic executing on an actual processor) that manipulates data values according to control signals (e.g., “commands,” “op codes,” “machine code,” etc.) and which produces corresponding output signals that are applied to operate a machine. A processor may, for example, be a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a digital signal processor (DSP), an ASIC, a radio-frequency integrated circuit (RFIC), or any combination thereof. A processor may further be a multi-core processor having two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously.

(20) In another context, a “Processor” (e.g., processor 514 in FIG. 5) is a company (often a third party) appointed to handle payment card (e.g., credit card, debit card) transactions. They have connections to various card networks and supply authorization and settlement services to merchants or payment service providers. In aspects, they can also move the money from an issuing bank to a merchant or acquiring bank.

(21) “Card Network” (or “Card Association”), in this context, refers to financial payment networks such as Visa®, MasterCard®, American Express®, Diners Club®, JCB®, and China Union-Pay®.

(22) “Acquiring Bank” or “Acquirer”, in this context, refers to a bank or financial institution that

accepts credit and/or debit card payments from affiliated card networks for products or services on behalf of a merchant or payment service provider.

(23) “Card Issuing Bank” or “Issuing Bank”, in this context, refers to a bank that offers card network or association-branded payment cards directly to consumers. An issuing bank assumes primary liability for the consumer's capacity to pay off debts they incur with their card.

(24) “Payment Information” includes information required to complete a transaction, and the specific type of information provided may vary by payment type. Some payment information will be sensitive (e.g., the card validation code), while other information might not be (e.g., zip code). For example, when a payment is made via a credit card or debit card, the payment information includes a primary account number (PAN) or credit card number, card validation code, and expiration month and year. In another payment example, made using an Automated Clearinghouse (ACH) transaction for example, the payment information includes a bank routing number and an account number within that bank.

(25) “Sensitive information” may not necessarily be related to payment information and may include other confidential personal information, such as medical (e.g., HIPAA) information, for example. The ambit of the term “Payment Information” includes “Sensitive Information” within its scope. In some examples, sensitive payment information may include “regulated payment information,” which may change over time. For example, currently a merchant cannot collect more than the first six (6) or the last four (4) numbers of a customer's PAN without generally needing to comply with Payment Card Industry (PCI) regulations. But card number lengths may change, and when they do, the “6 and 4” rules will likely change with them. These potential future changes are incorporated within the ambit of “regulated payment information,” which is, in turn, included within the ambit of the term “payment information” as defined herein.

(26) “Merchant”, in this context, refers to an entity that is associated with selling or licensing products and/or services over electronic systems such as the Internet and other computer networks. The merchant may be the direct seller/licensor, or the merchant may be an agent for a direct seller/licensor. For example, entities such as Amazon® sometimes act as the direct seller/licensor, and sometimes act as an agent for a direct seller/licensor.

(27) “Merchant Site”, in this context, refers to an e-commerce site or portal (e.g., website, or mobile app) of the merchant. In some figures, the merchant (e.g., a merchant **502** of FIG. 5) and merchant servers (e.g., merchant servers **506** of FIG. 5) are associated with the merchant site. The merchant site is associated with a client-side application and a server-side application. In one example embodiment, the merchant site includes the merchant servers **506** of FIG. 5, and the server-side application executes on the merchant servers **506**.

(28) “Payment Processor”, in this context, (e.g., a payment processor **510** in FIG. 5) refers to an entity or a plurality of entities that facilitate a transaction, for example between a merchant and a customer's electronic device. With reference to a high-level description illustrated in FIG. 5, in some examples described more fully below, the payment processor includes selected functionality of both the payment processor **510** and processor **514**/card networks **516**. For example, the payment processor **510** creates tokens and maintains and verifies publishable (non-secret) keys and secret keys. In the illustrated example, the processor **514**/card networks **516** are involved in authorizing or validating payment information. In one example embodiment, the payment processor **510** and the processor **514**/card networks **516** function together to authorize and validate payment information, issue a token, and settle any charges that are made. Accordingly, in this embodiment, “payment processor” refers to the functionality of the payment processor **510** and the functionality of the processor **514**/card networks **516**. In another example embodiment, wherein step (3) in the high-level description is not performed, and the payment processor **510** performs its own verification before issuing a token, the processor **514**/card networks **516** are still used for settling any charges that are made, as described in step (7). Accordingly, in this embodiment, “payment processor” may refer only to the functionality of the payment processor **510** with respect

to issuing tokens. Further, in the example arrangement shown, the payment processor **510**, the processor **514**, and the card networks **516** are shown as separate entities. In some examples, their respective functions may be performed by two entities, or even just one entity, with the entities themselves being configured accordingly.

(29) “Native Application” or “native app”, in this context, refers to an app commonly used with a mobile device, such as a smartphone or tablet. When used with a mobile device, the native app is installed directly onto the mobile device. Mobile device users typically obtain these apps through an online store or marketplace, such as an app store (e.g., Apple’s App Store, Google Play store). More generically, a native application is designed to run in the computer environment (machine language and operating system) that it is being run in. It can be referred to as a “locally installed application.” A native application differs from an interpreted application, such as a Java applet, which requires interpreter software. A native application also differs from an emulated application that is written for a different platform and converted in real time to run, and a web application that is run within the browser.

(30) The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative embodiments of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the inventive subject matter. It will be evident, however, to those skilled in the art, that embodiments of the inventive subject matter may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail. In this specification, the terms “user”, “cardholder”, and “consumer” are used interchangeably unless the context indicates otherwise.

(31) With reference to FIG. **1**, an example embodiment of a high-level SaaS network architecture **100** is shown. A networked system **116** provides server-side functionality via a network **110** (e.g., the Internet or a WAN) to a client device **108**. A web client **102** and a programmatic client, in the example form of a client application **104**, are hosted and execute on the client device **108**. The networked system **116** includes an application server **122**, which in turn hosts a publication system **106** (such as the publication system hosted at <https://stripe.com> by Stripe, Inc. of San Francisco, CA (herein “Stripe”) as an example of a payment processor **510**) that provides a number of functions and services to the client application **104** that accesses the networked system **116**. The client application **104** also provides a number of interfaces described herein, which present output of push payment decision routing to a user of the client device **108**.

(32) The client device **108** enables a user to access and interact with the networked system **116** and, ultimately, the publication system **106**. For instance, the user provides input (e.g., touch screen input or alphanumeric input) to the client device **108**, and the input is communicated to the networked system **116** via the network **110**. In this instance, the networked system **116**, in response to receiving the input from the user, communicates information back to the client device **108** via the network **110** to be presented to the user.

(33) An API server **118** and a web server **120** are coupled, and provide programmatic and web interfaces respectively, to the application server **122**. The application server **122** hosts the publication system **106**, which includes components or applications described further below. The application server **122** is, in turn, shown to be coupled to a database server **124** that facilitates access to information storage repositories (e.g., a database **126**). In an example embodiment, the database **126** includes storage devices that store information accessed and generated by the publication system **106**.

(34) Additionally, a third-party application **114**, executing on one or more third-party servers **112**, is shown as having programmatic access to the networked system **116** via the programmatic interface provided by the API server **118**. For example, the third-party application **114**, using information retrieved from the networked system **116**, may support one or more features or functions on a

website hosted by a third party.

(35) Turning now specifically to the applications hosted by the client device **108**, the web client **102** may access the various systems (e.g., the publication system **106**) via the web interface supported by the web server **120**. Similarly, the client application **104** (e.g., an “app” such as a payment processor app) accesses the various services and functions provided by the publication system **106** via the programmatic interface provided by the API server **118**. The client application **104** may be, for example, an “app” executing on the client device **108**, such as an iOS or Android OS application to enable a user to access and input data on the networked system **116** in an offline manner and to perform batch-mode communications between the programmatic client application **104** and the networked system **116**.

(36) Further, while the SaaS network architecture **100** shown in FIG. 1 employs a client-server architecture, the present inventive subject matter is, of course, not limited to such an architecture, and could equally well find application in a distributed, or peer-to-peer, architecture system, for example. The publication system **106** could also be implemented as a standalone software program, which does not necessarily have networking capabilities.

(37) FIG. 2 is a block diagram showing architectural details of a publication system **106**, according to some example embodiments. Specifically, the publication system **106** is shown to include an interface component **202** by which the publication system **106** communicates (e.g., over a network **110**) with other systems within the SaaS network architecture **100**.

(38) The interface component **202** is communicatively coupled to a payment processor component **204** that operates to provide push payment decision routing for a payment processor **510** in accordance with the methods described herein with reference to the accompanying drawings.

(39) FIG. 3 is a block diagram illustrating an example software architecture **306**, which may be used in conjunction with various hardware architectures herein described. FIG. 3 is a non-limiting example of a software architecture **306**, and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **306** may execute on hardware such as a machine **400** of FIG. 4 that includes, among other things, processors **404**, memory/storage **406**, and input/output (I/O) components **418**. A representative hardware layer **350** is illustrated and can represent, for example, the machine **400** of FIG. 4. The representative hardware layer **350** includes a processor **352** having associated executable instructions **304**. The executable instructions **304** represent the executable instructions of the software architecture **306**, including implementation of the methods, components, and so forth described herein. The hardware layer **350** also includes memory and/or storage modules as memory/storage **354**, which also have the executable instructions **304**. The hardware layer **350** may also comprise other hardware **356**.

(40) In the example architecture of FIG. 3, the software architecture **306** may be conceptualized as a stack of layers where each layer provides particular functionality. For example, the software architecture **306** may include layers such as an operating system **302**, libraries **322**, frameworks/middleware **318**, applications **316**, and a presentation layer **314**. Operationally, the applications **316** and/or other components within the layers may invoke API calls **308** through the software stack and receive a response as messages **312** in response to the API calls **308**. The layers illustrated are representative in nature, and not all software architectures have all layers. For example, some mobile or special-purpose operating systems may not provide a frameworks/middleware **318**, while others may provide such a layer. Other software architectures may include additional or different layers.

(41) The operating system **302** may manage hardware resources and provide common services. The operating system **302** may include, for example, a kernel **320**, services **324**, and drivers **326**. The kernel **320** may act as an abstraction layer between the hardware and the other software layers. For example, the kernel **320** may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services **324**

may provide other common services for the other software layers. The drivers **326** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **326** include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

(42) The libraries **322** provide a common infrastructure that is used by the applications **316** and/or other components and/or layers. The libraries **322** provide functionality that allows other software components to perform tasks in an easier fashion than by interfacing directly with the underlying operating system **302** functionality (e.g., kernel **320**, services **324**, and/or drivers **326**). The libraries **322** may include system libraries **344** (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematical functions, and the like. In addition, the libraries **322** may include API libraries **346** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as MPEG4, H.264, MP3, AAC, AMR, JPG, and PNG), graphics libraries (e.g., an OpenGL framework that may be used to render 2D and 3D graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **322** may also include a wide variety of other libraries **348** to provide many other APIs to the applications **316** and other software components/modules.

(43) The frameworks/middleware **318** provide a higher-level common infrastructure that may be used by the applications **316** and/or other software components/modules. For example, the frameworks/middleware **318** may provide various graphic user interface (GUI) functions **342**, high-level resource management, high-level location services, and so forth. The frameworks/middleware **318** may provide a broad spectrum of other APIs that may be utilized by the applications **316** and/or other software components/modules, some of which may be specific to a particular operating system **302** or platform.

(44) The applications **316** include built in applications **338** and/or third-party applications **340**. Examples of representative built in applications **338** may include, but are not limited to, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, and/or a game application. The third-party applications **340** may include any application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform and may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or other mobile operating systems. The third-party applications **340** may invoke the API calls **308** provided by the mobile operating system (such as the operating system **302**) to facilitate functionality described herein.

(45) The applications **316** may use built-in operating system functions (e.g., kernel **320**, services **324**, and/or drivers **326**), libraries **322**, and frameworks/middleware **318** to create user interfaces to interact with users of the system. Alternatively, or additionally, in some systems, interactions with a user may occur through a presentation layer, such as the presentation layer **314**. In these systems, the application/component “logic” can be separated from the aspects of the application/component that interact with a user.

(46) Some software architectures use virtual machines. In the example of FIG. 3, this is illustrated by a virtual machine **310**. The virtual machine **310** creates a software environment where applications/components can execute as if they were executing on a hardware machine (such as the machine **400** of FIG. 4, for example). The virtual machine **310** is hosted by a host operating system (e.g., the operating system **302** in FIG. 3) and typically, although not always, has a virtual machine monitor **358**, which manages the operation of the virtual machine **310** as well as the interface with the host operating system (e.g., the operating system **302**). A software architecture executes within the virtual machine **310** such as an operating system (OS) **336**, libraries **334**, frameworks **332**, applications **330**, and/or a presentation layer **328**. These layers of software architecture executing

within the virtual machine **310** can be the same as corresponding layers previously described or may be different.

(47) FIG. **4** is a block diagram illustrating components of a machine **400**, according to some example embodiments, able to read instructions **304** from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **4** shows a diagrammatic representation of the machine **400** in the example form of a computer system, within which instructions **410** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **400** to perform any one or more of the methodologies discussed herein may be executed. As such, the instructions **410** may be used to implement modules or components described herein. The instructions **410** transform the general, non-programmed machine **400** into a particular machine **400** programmed to carry out the described and illustrated functions in the manner described. In alternative embodiments, the machine **400** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **400** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **400** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **410**, sequentially or otherwise, that specify actions to be taken by the machine **400**. Further, while only a single machine **400** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **410** to perform any one or more of the methodologies discussed herein.

(48) The machine **400** may include processors **404** (including processor **408** and processor **412**), memory/storage **406**, and I/O components **418**, which may be configured to communicate with each other such as via a bus **402**. The memory/storage **406** may include a memory **414**, such as a main memory, or other memory storage, and a storage unit **416**, both accessible to the processors **404** such as via the bus **402**. The storage unit **416** and memory **414** store the instructions **410** embodying any one or more of the methodologies or functions described herein. The instructions **410** may also reside, completely or partially, within the memory **414**, within the storage unit **416**, within at least one of the processors **404** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **400**. Accordingly, the memory **414**, the storage unit **416**, and the memory of the processors **404** are examples of machine-readable media.

(49) The I/O components **418** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **418** that are included in a particular machine **400** will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **418** may include many other components that are not shown in FIG. **4**. The I/O components **418** are grouped according to functionality merely for simplifying the following discussion, and the grouping is in no way limiting. In various example embodiments, the I/O components **418** may include output components **426** and input components **428**. The output components **426** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **428** may include alphanumeric input

components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(50) In further example embodiments, the I/O components **418** may include biometric components **430**, motion components **436**, environment components **434**, or position components **438**, among a wide array of other components. For example, the biometric components **430** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **436** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environment components **434** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas sensors to detect concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **438** may include location sensor components (e.g., a Global Positioning System (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(51) Communication may be implemented using a wide variety of technologies. The I/O components **418** may include communication components **440** operable to couple the machine **400** to a network **432** or devices **420** via a coupling **424** and a coupling **422**, respectively. For example, the communication components **440** may include a network interface component or other suitable device to interface with the network **432**. In further examples, the communication components **440** may include wired communication components, wireless communication components, cellular communication components, near field communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **420** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

(52) Moreover, the communication components **440** may detect identifiers or include components operable to detect identifiers. For example, the communication components **440** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **440**, such as location via Internet Protocol (IP) geo-location, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

(53) In some embodiments, a JavaScript library can be embedded into a merchant **502**'s checkout

form to handle credit card information. When a user attempts to complete a transaction using the checkout form, it sends the credit card information directly from the user's browser to the payment processor **510**'s servers. The JavaScript library provides merchants **502** with a set of technologies that can be easily and quickly integrated to securely accept payments online. With the JavaScript library, merchants **502** retain full control of their customers' payment flows, but their servers are never exposed to sensitive payment information.

(54) When added to a merchant's payment form, the JavaScript library automatically intercepts the payment form submission, sending payment information directly to the payment processor **524** and converting it to a single-use token. The single-use token can be safely passed to the merchant's systems and used later to charge customers. Merchants **502** have complete control of their customers' payment experience without ever handling, processing, or storing sensitive payment information.

(55) Viewed generally in one example, and with reference to FIG. 5, a payment processing flow is now described:

(56) The merchant's customer **504** uses an Internet-enabled browser **512** to visit the merchant's site. The customer **504** is served a JavaScript library-enabled payment form **520** using standard web technologies. The customer **504** enters the specified information including their payment information **524** and submits the payment form **520**. The billing info portion of the payment form **520** is for payment via a credit card or debit card. If payment is to be made via an ACH transaction, the billing info portion of the payment form **520** will request a bank routing number and an account number within that bank, and possibly additional information, such as the bank name and whether the account is a checking or savings account.

(57) The customer's payment information **524** is sent from the customer's browser **512** to the payment processor **510**, never touching the merchant servers **506**. In this manner, the client-side application electronically sends payment information **524** retrieved from the customer's electronic device to the payment processor **510**. The client-side application does not send the payment information **524** to the server-side application.

(58) In one preferred embodiment, the payment processor **510** submits the relevant transaction to a processor **514** or directly to the card network **516** for authorization or validation of the payment information **524**. The card network **516** sends the request to the card issuing bank **518**, which authorizes the transaction. In this embodiment, the payment processor **510** and the processor **514**/card network **516** function together as a payment processor **510**. In another example embodiment, this step is performed without any communication to the processor **514**/card network **516**. Instead, the payment processor **510** performs its own authorization or validation of the payment information **524** using heuristic means, such as by checking the bank identification number (BIN), also referred to as the issuer identification number (IIN), against a database **126** of known, valid BINs on file with the payment processor **510**. (The BIN is a part of the bank card number, namely the first six digits.) In yet another example embodiment, this step is not performed at all since the authorization or validation is not necessary for the next step (4) to succeed. That is, it is acceptable to create a single-use token in step (4) that represents payment information **524** which has not been validated in any way.

(59) If authorized, the payment processor **510** will generate and return a secure, single-use token **522** to the customer's browser **512** that represents the customer's payment information **524** but does not leak any sensitive information. In the example embodiment wherein step (3) is not performed, the payment processor **510** performs this step without waiting to receive authorization from the processor **514** or the card network **516**. In this manner, the payment processor **510** creates the token **522** from the payment information **524** sent by the client-side application, wherein the token **522** functions as a proxy for the payment information **524**.

(60) The payment form **520** is submitted to the merchant servers **506**, including the single-use token **522**. More specifically, the payment processor **510** sends the token **522** to the client-side

application, which, in turn, sends the token **522** to the server-side application for use by the server-side application in conducting the transaction.

(61) The merchant **502** uses the single-use token **522** to submit a charge request to the payment processor **510** (or to create a customer object for later use). In this step, the payment processor **510** submits a request to authorize the charge to the processor **514** or directly to the card network **516**. This authorization specifies the actual amount to charge the credit card. If an authorization was already done in step (3) for the correct amount, this authorization request can be skipped. This may be a one-time payment for a merchant item, or it may involve registering the payment information **524** with the merchant site for subsequent use in making a payment for a merchant item (a so-called “card on file” scenario). Using the process described in steps (1) through (6), the payment information **524** can be used by the server-side application via the token **522** without the server-side application being exposed to the payment information **524**.

(62) The payment processor **510** settles the charge on behalf of the merchant **502** with the processor **514** or directly with the card network **516**.

(63) The card network **516** causes the funds to be paid by the card issuing bank **518** to the payment processor **510** or to the payment processor's acquiring bank **508**.

(64) The payment processor **510** causes the settled funds to be sent to the merchant **502** (or to the merchant's bank **526**), net of any applicable fees.

(65) The card issuing bank **518** collects the paid funds from the customer **504**.

(66) Not all of the steps listed above need happen in real time. Other examples, arrangements, and functionality are possible. Applicant's published patent application US 2013/0117185 A1 is incorporated by reference in its entirety in this regard. Typically, when the merchant's customer **504** submits the payment form **520** in step (1), steps (1) through (6) happen in real time and steps (7) through (10) happen later, usually once per day, as a batch process settling all of the funds for all of the payment processor **510**'s merchants **502**. In some examples, the payment processor **510** uses an HTTP-based tokenization API in steps (2) and (4) above. Some broader examples may be considered as “tokenization as a service,” in which any data is tokenized. One general example may facilitate a merger and acquisition (M&A) analysis in which companies want to compare an overlap in their customer bases. A payment processor **510** (acting as a tokenization service) can tokenize the customers **504** of each company and compare the overlap without revealing confidential information to either party. Unique payment tokens can be adapted to enable and facilitate such a tokenization service.

(67) The As mentioned above, factors that can present technical challenges include resolving the location of a user's electronic device so that a location-dependent calculation or determination can be performed with accurate location data.

(68) Thus, in some embodiments, a system is provided for resolving the location of user's electronic device and then performing a location-dependent calculation. An example system may comprise a network; one or more hardware processors; and a memory storing instructions that, when executed by at least one processor among the processors, cause the system to perform operations comprising, at least: determining a geographic location of a user's electronic device using first data; in response to determining the determined geographic location is insufficient for performing a location-dependent calculation, requesting second data to determine an updated geographic location of the user's electronic device; performing, with an API, the location-dependent calculation based on the updated geographic location; and completing, with the API, a transaction using the location-dependent calculation.

(69) FIG. 6 illustrates a system **600** in accordance with one embodiment. The system **600** includes a server **614** having a Product Catalog **602** for a merchant, such as merchant **502**, and a Payment Intent API **604**. The Product Catalog **602** includes products a merchant sells (available for purchase). These products will be attached to individual transaction line items in a customer basket at a user device **616**, which is communicatively coupled to the server **614**, to inform Payment

Intent API **604** of how to treat them for tax purposes. Some product types have reduced tax rates, such as ebooks in most of Europe. Products with no tax code use a default from merchant settings—either fully taxable or not taxable. Example tax codes to apply to products include:

(70) TABLE-US-00001 TAX CODE DESCRIPTION DM201010 Digital Media > ebooks > Less Than Permanent Use Rights Digital Goods & Services DM201020 Digital Media > ebooks > Permanent Use Rights Digital Goods & Services DM201030 Digital Media > ebooks > Education & Reference Digital Goods & Services

(71) The Payment Intent API **604** comprises Tax Rates **606**, Location Determination **608**, Tax Determination **610** and Payment Processing **612**. In an embodiment, the system **600** can include the payment processor **510**. The Payment Intent API **604** creates an object that represents a merchant's intent to collect payment from a user of the user device **616** and tracks the lifecycle of the payment process through each stage.

(72) As will be described in further detail below, the Location Determination **608** determines a location of the user device **616**. The Tax Determination **610** performs a location dependent calculation, such as calculating sales tax using the determined location and applying Tax Rates **606** to a product in the Product Catalog **602** based on tax code and/or location. The Payment Processing **612** then completes a transaction for the taxed product as discussed in conjunction with FIG. 5 or other methods. For example, displaying the final (tax) calculation on the user device **616** and accepting payment input from the user device **616**. Accordingly, both tax calculation and payment can be performed with a call to a single API instead of to multiple APIs to speed calculation and reduce communication traffic in the server **614** due to multiple API calls.

(73) The Payment Intent API **604** supports both tax inclusive (expected taxes built into pricing) and tax exclusive (taxes added after pricing) rates. While tax-inclusive merchants must still collect information about their end customer's location in order to accurately calculate rates, this information won't change the total amount that their customer pays, so merchants can provide tax later in a payment flow. Conversely, tax-exclusive merchants must collect information about their customer's location earlier in the flow so that the Payment Processing **612** can accurately calculate tax rates for the merchant to display to their end customer before the customer confirms the payment.

(74) In an example, the Payment Intent API **604** can include a TaxRate object having the following fields:

(75) TABLE-US-00002 Field Meaning inclusive Whether this tax is included in the subtotal price or not. Determined by the tax_behavior of the price object on each line item. jurisdiction Outside the US: The country charging the tax In the US/Canada: The state/province charging the tax percent The base percentage of the tax. This percentage may not correspond perfectly to the percentage charged, due to local tax rules. display_name A local display name for this tax. For certain taxes, this may be the legally mandated display name (for example French VAT taxes will appear “TVA”). For other taxes, this is a user-localized version of the tax name (e.g., a sales may appear as a “taxe de vente” to a French speaker). tax_type The type of the tax. country The country charging the tax state In the US/Canada: The state/province charging the tax Outside the US: Not present

(76) The Location Determination **608** determines location initially, in an example, by determining a location associated with an IP address of the user device **616**. This allows for estimating taxes with no customer friction (requiring no manual input from the customer). This can be passed to Payment Intent object at creation or by updating the Payment Intent object. However, IP addresses will not always provide enough information to perform a location dependent calculation, such as estimate taxes, so the Location Determination **608** may request a user's country, Postal Code or full address instead of or in addition. For example, an initial location determination (e.g., IP address or via other methods) may yield an area having different tax rates and therefore require a location determination with higher resolution to determine accurate tax. For example, a postal code for initial location

determination can yield several towns each with their own tax rate. An IP address may be insufficient because if it yields a location of a controlling agency that the user device **616** is connected to and not the true location of the user device **616**. For example, if the user device **616** is in the northern US, the IP address may show Canada, thereby requiring further resolution of the location of the user device **616**. Accordingly, an IP address indicated a physical location at a border region will require further resolution by the Location Determination **608**.

(77) In an example, in order to show taxes on a cart before a customer enters their payment information, the merchant may collect a subset of the billing address before collecting the Payment Method. To do this, the Location Determination **608** will prompt the user for billing information.

FIG. **8A** and FIG. **8B** show the transition from an IP address location determination at **802** to a user-entered billing address at **804** in a GUI **800**.

(78) In order to maintain clarity that this address is lower priority than the actual billing address and also not the correct place for collecting a full billing address when collecting payment information, the tax billing address allows for less detail than a full address. The Location Determination **608** will ask for a country, postal code and/or state/province in the tax billing address.

(79) Alternatively to using the IP address (e.g., if the IP address is insufficient for the location dependent calculation), the Location Determination **608** can request that the user device **616** send current location as determined by a global positioning system (e.g., GPS, GLONASS, etc.) reading by, for example, sending a request to the user device **616** to call a geolocation API and report back the result. In another example, the Location Determination **608** can determine current location based on a recent photograph. For example, the Location Determination **608** can find a recent photograph of the Eiffel Tower in memory of the user device **616**, search for that image in a database of images (on the server **614** or elsewhere) with correlated locations and determine that the device is in Paris, France. In an example, a user may wish to enter an address different from the location of the user device **616** (e.g., a shipping address different from a location of the user device **616**).

(80) If the tax amount on a payment changes (e.g., user changes tax location or the Location Determination **608** resolves location to be different than an initially determined location), the Payment Intent API **604** may need to reconfirm the payment by displaying the updated calculation on the user device **616** and asking the user to confirm again. When reconfirming a payment, the Payment Intent API **604** may allow the user to change the Payment Method and billing address, in which case the tax amount may change again, requiring further reconfirmation.

(81) In another example, the Payment Intent API **604** can be employed to provide access to a computer network based on resolved location. For example, the Location Determination **608** determines location and the Payment Processing **612** would grant access (authorization session) based on the location (e.g., access if in the US and no access if in a foreign country; access if at a specific building, limited access otherwise; authorized session for a specific amount of time based on location, etc.). In another example, the Payment Intent API **604** can be employed to complete a transaction that transfers fungible goods such as electricity or oil. For example, electricity could be provided to a building in which the user device **616** is located.

(82) FIG. **7** illustrates a method **700** in accordance with one embodiment. After a user of a mobile device has selected an item for purchase, the method **700** determines (**702**) the tax code for the item based on an entry in the Product Catalog **602**. If (**704**) the item is not taxable the method **700** ends. Otherwise, the location is determined (**706**) via a first method as discussed above. For example, it initially may be via IP address of the user device **616**. If (**708**) the determined location is insufficient to perform a location-dependent calculation (e.g., sales tax calculation) or determination (e.g., network access), the location is resolved (**710**) by using a second method to determine location as discussed above. This may include, for example, requesting a user of the device to enter a location or requesting the user's device to send its location based on a read of a

GPS device embedded within the user device. Next, a calculation/determination is performed (712) based on the location. For example, a sales tax calculation can be calculated based on location and tax code indicating how the item should be taxed. Further, for example, the determination could be to grant access to a network based on location. Alternatively, for example, the item could include electricity and the determining based on location could determine if the location can receive electricity and/or what type of electricity (e.g., solar, wind, conventional, etc.). Next, a tax breakout is displayed (714) and the transaction completed (716), which can include processing payment for the item, transmitting electricity, granting access to a network, etc.

(83) Examples include:

(84) 1. A method for location-dependent determination, comprising: determining, with one or more processors, a geographic location of a user's electronic device using first data; in response to determining the determined geographic location is insufficient for performing a location-dependent determination, requesting second data to determine an updated geographic location of the user's electronic device; performing, with an Application Programming Interface (API), the location-dependent determination based on the updated geographic location; and completing, with the API, a transaction using the location-dependent determination.

(85) 2. The method of example 1, wherein the first data is an Internet Protocol (IP) address of the user's electronic device and the second data is a postal code entered by a user into the electronic device.

(86) 3. The method of any of the preceding examples, wherein the requesting requests the electronic device's location based on a global positioning system reading.

(87) 4. The method of any of the preceding examples, wherein the requesting requests the user to enter a shipping postal code.

(88) 5. The method of any of the preceding examples, wherein the performing the location-dependent determination is further based on a tax code of an item to be purchased in the transaction.

(89) 6. The method of any of the preceding examples, further comprising generating a product catalog including tax codes for items available for purchase.

(90) 7. The method of any of the preceding examples, wherein the transaction includes a purchase of an item and the location-dependent determination includes tax associated with the purchase.

(91) 8. The method of any of the preceding examples, wherein the location-dependent determination includes calculating an authorization session to access a computer network and the completing includes granting access to the electronic device to access the computer network for the session.

(92) 9. The method of any of the preceding examples, wherein the location-dependent determination includes calculating a type of electricity to transmit to a location of the user's electronic device and the completing includes transmitting the electricity to the determined location.

(93) 10. A non-transitory machine-readable medium comprising instructions which, when read by a machine, cause the machine to perform operations, the operations comprising, at least: determining a geographic location of a user's electronic device using first data; in response to determining the determined geographic location is insufficient for performing a location-dependent determination, requesting second data to determine an updated geographic location of the user's electronic device; performing, with an Application Programming Interface (API), the location-dependent determination based on the updated geographic location; and completing, with the API, a transaction using the location-dependent determination.

(94) 11. A computing apparatus for location-dependent determinations, the computing apparatus comprising: one or more hardware processors; and a memory storing instructions that, when executed by at least one processor among the processors, cause the system to perform operations comprising, at least: determining a geographic location of a user's electronic device using first data;

in response to determining the determined geographic location is insufficient for performing a location-dependent determination, requesting second data to determine an updated geographic location of the user's electronic device; performing, with an Application Programming Interface (API), the location-dependent determination based on the updated geographic location; and completing, with the API, a transaction using the location-dependent determination.

(95) 12. The computing apparatus of any of the preceding examples, wherein the first data is an Internet Protocol (IP) address of the user's electronic device and the second data is a postal code entered by a user into the electronic device.

(96) 13. The computing apparatus of any of the preceding examples, wherein the requesting requests the electronic device's location based on a global positioning system reading.

(97) 14. The computing apparatus of any of the preceding examples, wherein the requesting requests the user to enter a shipping postal code.

(98) 15. The computing apparatus of any of the preceding examples, wherein the performing the location-dependent determination is further based on a tax code of an item to be purchased in the transaction.

(99) 16. The computing apparatus of any of the preceding examples, wherein the instructions further configure the apparatus to generate a product catalog including tax codes for items available for purchase.

(100) 17. The computing apparatus of any of the preceding examples, wherein the transaction includes the purchase of an item and the location-dependent determination includes tax.

(101) 18. The computing apparatus of any of the preceding examples, wherein the location-dependent determination includes calculate an authorization session to access a computer network and the completing includes granting access to the electronic device to access the computer network for the session.

(102) 19. The computing apparatus of any of the preceding examples, wherein the location-dependent determination includes calculating an amount of time for the authorization session.

(103) 20. The computing apparatus of any of the preceding examples, wherein the location-dependent determination includes calculate a type of electricity to transmit to a location of the user's electronic device and the completing includes transmitting the electricity to the determined location.

(104) Although the subject matter has been described with reference to specific example embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader scope of the disclosed subject matter.

Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. The accompanying drawings that form a part hereof show by way of illustration, and not of limitation, specific embodiments in which the subject matter may be practiced. The embodiments illustrated are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed herein. Other embodiments may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. This Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by any appended claims, along with the full range of equivalents to which such claims are entitled.

(105) Such embodiments of the inventive subject matter may be referred to herein, individually and/or collectively, by the term "invention" merely for convenience and without intending to voluntarily limit the scope of this application to any single invention or inventive concept if more than one is in fact disclosed. Thus, although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above

embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description.

Claims

1. A method comprising: receiving, by one or more processors, payment information for each of a plurality of users from a plurality of client-side applications associated with the plurality of users; generating, by the one or more processors, a token for each of the plurality of users from the payment information sent by the plurality of client-side applications, each token configured to be used to complete a transaction through the one or more processors; comparing, by the one or more processors, a first set of tokens generated from payment information of a first subset of the plurality of users for a first entity with a second set of tokens generated from payment information of a second subset of the plurality of users for a second entity; and responsive to identifying at least one matching token between the first set of tokens and the second set of tokens, generating, by the one or more processors, a record comprising a list of matching tokens that are included in each of the first set of tokens and the second set of tokens and represent a third subset of the plurality of users that are common to the first entity and second entity.
2. The method of claim 1, wherein generating the token for each of the plurality of users from the payment information sent by the plurality of client-side applications comprises generating, by the one or more processors, the token using a Hypertext Transfer Protocol (HTTP)-based tokenization application programming interface.
3. The method of claim 1, comprising: receiving, by the one or more processors, a first token from a server-side application; and settling, by the one or more processors, a first transaction based on the first token.
4. The method of claim 1, wherein a first client-side application of the plurality of client-side applications transmits a first token to a server-side application to complete a transaction comprising registering payment information of the first token with a first merchant website for subsequent use in making a payment for a merchant item.
5. The method of claim 1, wherein a first client-side application of the plurality of client-side applications executes within a browser executing on a computing device.
6. The method of claim 1, further comprising: determining, by the one or more processors, a geographic location of a first user's electronic device using first data, the first user's electronic device executing a first client-side application to complete a first transaction; performing, by the one or more processors via an Application Programming Interface (API), a location-dependent determination based on the geographic location; and completing, by the one or more processors via the API, the first transaction using the location-dependent determination based on a first token generated by the one or more processors from payment information of the first user.
7. The method of claim 1, further comprising: enforcing, by the one or more processors, a rule that prevents a previously used token from being used for another transaction.
8. The method of claim 1, comprising: authorizing, by the one or more processors, a first transaction, wherein generating a first token for a first user of the plurality of users comprises generating, by the one or more processors, the first token responsive to authorizing the first transaction.
9. The method of claim 8, wherein authorizing the first transaction comprises: transmitting, by the one or more processors, data of the first transaction to a first remote computing device; and receiving, by the one or more processors, authorization for the first transaction from the first remote computing device or a second remote computing device.
10. The method of claim 8, wherein authorizing the first transaction comprises: executing, by the one or more processors, a rule using data of a transaction card of the payment information as input; and determining, by the one or more processors, the first transaction is authorized based on the

executing the rule.

11. A system comprising: one or more processors coupled with a non-transitory memory and configured to: receive payment information for each of a plurality of users from a plurality of client-side applications associated with the plurality of users; generate a token for each of the plurality of users from the payment information sent by the plurality of client-side applications, each token configured to be used to complete a transaction through the one or more processors; compare a first set of tokens generated from payment information of a first subset of the plurality of users for a first entity with a second set of tokens generated from payment information of a second subset of the plurality of users for a second entity; and responsive to identifying at least one matching token between the first set of tokens and the second set of tokens, generate a record comprising a list of matching tokens that are included in each of the first set of tokens and the second set of tokens and represent a third subset of the plurality of users that are common to the first entity and second entity.

12. The system of claim 11, wherein the one or more processors are configured to generate the token for each of the plurality of users from the payment information sent by the plurality of client-side applications by generating the token using a Hypertext Transfer Protocol (HTTP)-based tokenization application programming interface.

13. The system of claim 11, wherein the one or more processors are further configured to: receive a first token from a server-side application; and settle a first transaction based on the first token.

14. The system of claim 11, wherein a first client-side application of the plurality of client-side applications transmits a first token to a server-side application to complete a first transaction comprising purchasing an item from a first merchant website associated with the first client-side application and the server-side application.

15. The system of claim 11, wherein a first client-side application of the plurality of client-side applications transmits a first token to a server-side application to complete a first transaction comprising registering payment information of the first token with a first merchant website for subsequent use in making a payment for a merchant item.

16. The system of claim 11, wherein the one or more processors are further configured to: authorize a first transaction, wherein the one or more processors are further configured to generate a first token for a first user of the plurality of users by generating the first token responsive to authorizing the first transaction.

17. A non-transitory machine-readable medium comprising instructions which, when executed by one or more processors, cause the one or more processors to: receive payment information for each of a plurality of users from a plurality of client-side applications associated with the plurality of users; generate a token for each of the plurality of users from the payment information sent by the plurality of client-side applications, each token configured to be used to complete a transaction through the one or more processors; compare a first set of tokens generated from payment information of a first subset of the plurality of users for a first entity with a second set of tokens generated from payment information of a second subset of the plurality of users for a second entity; and responsive to identifying at least one matching token between the first set of tokens and the second set of tokens, generate a record comprising a list of matching tokens that are included in each of the first set of tokens and the second set of tokens and represent a third subset of the plurality of users that are common to the first entity and second entity.

18. The non-transitory machine-readable medium of claim 17, wherein executing the instructions further causes the one or more processors to: receive a first token from a first server-side application; and settle a first transaction based on the first token.

19. The non-transitory machine-readable medium of claim 17, wherein a first client-side application of the plurality of client-side applications transmits a first token to a server-side application to complete a first transaction comprising registering payment information of the first token with a first merchant website for subsequent use in making a payment for a merchant item.

20. The non-transitory machine-readable medium of claim 17, wherein executing the instructions further causes the one or more processors to: authorize a first transaction, wherein executing the instructions causes the one or more processors to generate a first token for a first user of the plurality of users by generating the first token responsive to authorizing the first transaction.
